Individual Machine Learning Tutorial

Title: Understanding Random Forests: How the Number of Trees Affects Classification Performance

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GitHub Repository: https://github.com/mn25abw/random-forest-tutorial

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1. Introduction

Machine learning algorithms are widely used to solve classification problems in areas such as healthcare, cybersecurity, finance, and recommendation systems. One of the most popular and effective classification algorithms is Random Forest. Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Decision trees are simple and interpretable models, but they often suffer from overfitting when trained on complex datasets. Random Forest addresses this problem by creating many decision trees and combining their predictions using a voting mechanism. This generally produces more stable and reliable predictions than a single tree.

The purpose of this tutorial is to investigate how changing the number of trees in a Random Forest model affects classification performance. The tutorial demonstrates the technique using the Breast Cancer Wisconsin dataset available in Scikit-learn. Several Random Forest models were trained using different numbers of trees, and their performance was evaluated using classification metrics and visualisations.

This tutorial also discusses ethical considerations associated with machine learning in healthcare applications.

2. What is Random Forest?

Random Forest is a supervised machine learning algorithm that is commonly used for both classification and regression problems. It is part of a group of techniques known as ensemble learning methods. Ensemble learning works by combining multiple machine learning models together in order to improve overall prediction performance and reliability.

The Random Forest algorithm is made up of many individual decision trees. Instead of relying on a single tree, the model creates a collection, or "forest," of trees that work together to make predictions. Each decision tree is trained using a randomly selected subset of the training data through a method called bootstrap sampling. In addition, when splitting data at each node, the algorithm only considers a random subset of the available features. These randomisation processes help create diversity among the trees, which improves the overall strength of the model.

When the model is used for prediction, each decision tree produces its own output. For classification tasks, the final prediction is determined using majority voting, where the class predicted by most trees becomes the final output. This collaborative approach helps Random Forest achieve better accuracy and stability compared to a single decision tree.

One of the main advantages of Random Forest is its high classification accuracy. Because the model combines many trees, it is generally more reliable and less likely to overfit the training data than an individual decision tree. Random Forest can also handle datasets with a large number of features and performs well even when some data contains noise or missing values. Another useful feature of the algorithm is its ability to calculate feature importance, allowing users to identify which variables contribute most strongly to predictions.

Despite these advantages, Random Forest also has some limitations. Training large forests with many trees can increase computational cost and processing time. The model is also less interpretable than a single decision tree because it combines predictions from many different trees, making its decision-making process harder to explain. Additionally, larger Random Forest models may require more memory and storage resources.

3. Dataset

The tutorial uses the Breast Cancer Wisconsin dataset provided by Scikit-learn. This dataset contains medical diagnostic measurements used to classify breast tumours as malignant or benign.

The dataset contains:

- 569 samples
- 30 numerical input features
- Binary classification labels

Examples of features include:

- Mean radius
- Mean texture
- Mean smoothness
- Mean compactness

The dataset was selected because it is well-structured, widely used for educational purposes, and suitable for demonstrating classification algorithms.

The target classes are:

- Malignant
- Benign

The dataset was loaded directly using the load_breast_cancer() function from Scikit-learn.

4. Experimental Setup

The dataset was divided into training and testing subsets using an 80/20 split. The training set was used to train the Random Forest models, while the testing set was used to evaluate model performance.

Several Random Forest classifiers were trained using different numbers of trees:

1 tree
5 trees
10 trees
50 trees
100 trees

The experiment aimed to observe how increasing the number of trees influences:

Training accuracy
Testing accuracy
Overfitting behaviour
Prediction stability

The following Python libraries were used:

NumPy
Pandas
Matplotlib
Scikit-learn

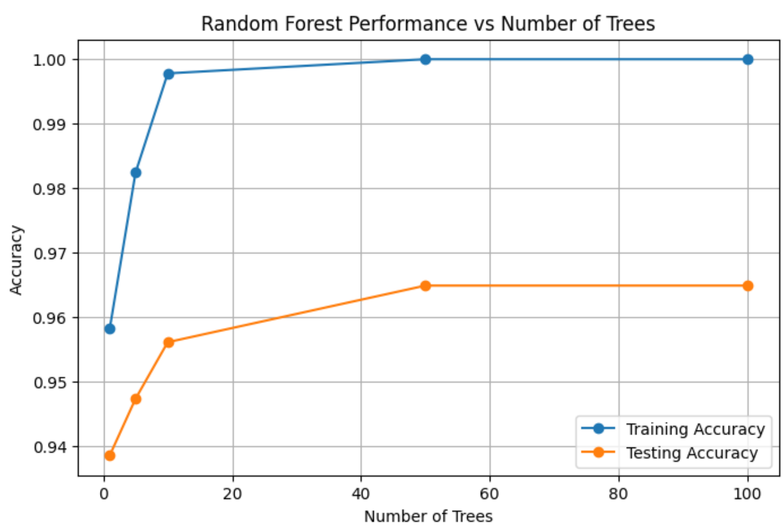
The model was implemented using the RandomForestClassifier class from Scikit-learn.

5. Results and Analysis

Accuracy Comparison

The results showed that testing accuracy generally improved as the number of trees increased.
The model with only one tree behaved like a standard decision tree and showed signs of overfitting. It had very high training accuracy, but lower testing accuracy.
The testing performance improved and became more stable with the increase of trees. This was due to the fact that the ensemble learning process helped to reduce the variance of the individual trees.
The best result was obtained with 100 trees.
The graph for accuracy showed that adding more trees did not do much to improve performance. This demonstrates diminishing returns, where increasing the number of trees eventually yields only marginal gains at increased computational expense.

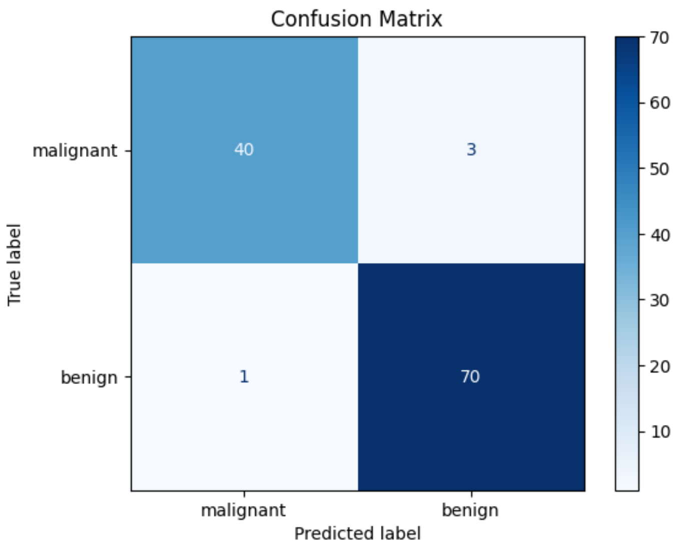
Figure 1



Confusion Matrix

A confusion matrix was created to evaluate the final Random Forest model trained with 100 trees.
The confusion matrix showed that the model was able to classify most of the benign and malignant tumour samples correctly. Only a small number of samples were misclassified.
These results demonstrate that Random Forest can achieve highly accurate classification performance on medical datasets.

Figure 2
Confusion matrix for the Random Forest classifier.

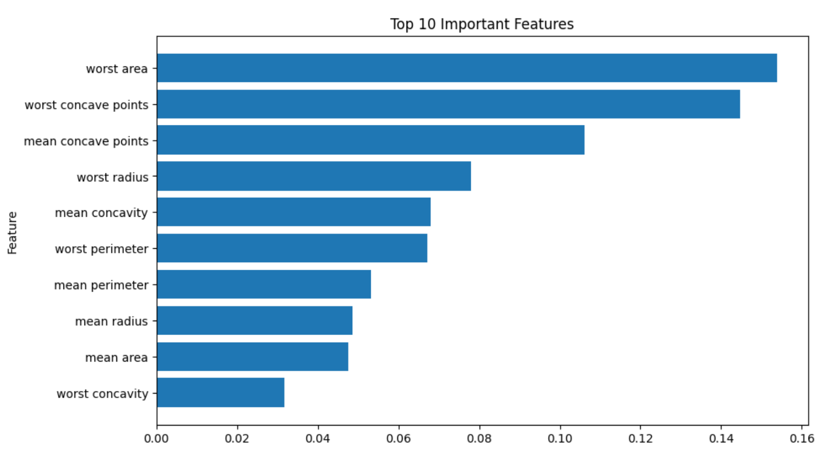


Feature Importance

One of the biggest advantages of Random Forest is that it can automatically calculate feature importance .
Feature importance analysis showed that several among the various tumour measurements contributed significantly to classification performance. The most important predictors included factors related to the tumour radius, texture and concavity.
Feature importance is useful because it helps researchers understand which variables have the greatest influence on the model decisions.

Figure 3

Top 10 most important features in the Random Forest model.



Limitations

Although Random Forest performs well on many classification tasks, it has several limitations. Large forests can increase computational time and memory usage, especially with large datasets. In addition, Random Forest models are less interpretable than simpler models such as individual decision trees, making it more difficult to explain predictions in sensitive applications such as healthcare.

Ethical Considerations

Random Forest models can be very accurate, but ethical concerns are still important, especially when they are used in healthcare.

An important problem is dataset bias. If the training data is not representative of all patient populations, the model may create biased predictions for under-represented groups. This can result in disparities in health outcomes

Explainability too is a concern. Random Forest models are harder to interpret than pure decision trees as they combine predictions from multiple trees. In healthcare domains, medical experts often need understandable explanations before they trust automated systems.

Inaccurate forecasts are another concern. While false positives may result in needless worry and medical treatments, false negatives could postpone diagnosis and treatment.

To improve ethical AI practice, machine learning systems should be:

- Carefully validated
- Trained using diverse datasets
- Monitored for bias
- Used alongside human expertise rather than replacing professionals

Conclusion

This tutorial demonstrated how the performance of a Random Forest classifier changes as the number of trees increases. The experiments showed that larger forests generally produced more accurate and stable predictions compared to a single decision tree.

Increasing the number of trees improved testing accuracy and reduced overfitting by combining predictions from multiple diverse models. However, the results also showed diminishing returns, where adding additional trees eventually produced only small improvements while increasing computational cost.

Feature importance analysis further demonstrated how Random Forest models can help identify the variables that contribute most strongly to predictions.

Overall, Random Forest remains one of the most effective and widely used machine learning algorithms for classification tasks due to its strong predictive performance, robustness, and flexibility.

References

Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), pp.5–32.

Scikit-learn Developers. *RandomForestClassifier Documentation*. Available at: Scikit-learn RandomForestClassifier Documentation

Scikit-learn Developers. *Breast Cancer Wisconsin Dataset Documentation*. Available at: Breast Cancer Dataset Documentation

Pedregosa, F. et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, pp.2825–2830.